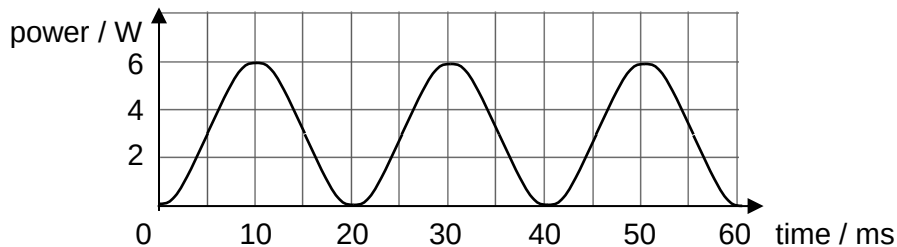


- 25** The figure below shows the power supplied by a sinusoidal a.c. source to a load of resistance $1.5\ \Omega$ over 60 ms.



Which of the following is correct?

- A** The mean value of the power supplied is 4.2 W.
- B** The peak value of the current supplied is 2.0 A.
- C** The root mean square value of the voltage supplied is 4.5 V
- D** The frequency of the source is 50 Hz.

Ans: B

$$\langle P \rangle = \frac{1}{2} \times P_0 = 3.0 \text{ W}$$

$$P_0 = I_0^2 R \rightarrow I_0 = 2.0 \text{ A}$$

$$\langle P \rangle = V_{rms}^2 / R \rightarrow V_{rms} = 3.0 \text{ V}$$

$$\text{Period of power} = \frac{1}{2} \times \text{Period of current} \rightarrow f = 1/0.040 = 25 \text{ Hz}$$